Weekly report

Natalia Trigo Tomasevich

October 19, 2026

This week

- Competition model

This week

- Competition model
- Oblivious equilibrium

This week

- Competition model
- Oblivious equilibrium
- Simulations

Elements of the model

- Two symmetric providers, indexed by $j \in \{A, B\}$
- Two products, indexed by $i \in \{1, 2\}$
- Success probabilities: $0 < p_1 < p_2 < 1$
- Costs: $c_1 < c_2$
- Revenue: $R > c_2$

Customer's beliefs

For the competition model, the customer observes the state x_A, x_B, where x_j is the fraction of successes over the interactions with provider j. The customer has a belied about each provider j

$$\theta_A \mid x_A \sim \text{Beta}(1 + \frac{x_A}{1-\rho}, 1 + \frac{1-x_A}{1-\rho}), \tag{1}$$

$$\theta_B \mid x_B \sim \text{Beta}(1 + \frac{x_B}{1-\rho}, 1 + \frac{1-x_B}{1-\rho}) \tag{2}$$

Customer's choice

The customer uses Thompson sampling (TS), drawing independently from the two belief distributions and choosing

$$y_t = \begin{cases} A, & \widetilde{\theta}_A > \widetilde{\theta}_B, \\ B, & \widetilde{\theta}_B > \widetilde{\theta}_A. \end{cases} \tag{3}$$

Thus the selection probabilities are

$$\begin{aligned} D_A(x_A, x_B) &= \mathbb{P}(\widetilde{\theta}_A > \widetilde{\theta}_B \mid x_A, x_B), \\ D_B(x_A, x_B) &= 1 - D_A(x_A, x_B). \end{aligned} \tag{4}$$

State transitions

Superscripts s and f denote success and failure, respectively. The state transitions are

$$\begin{aligned} T_A^s(x) &= (\rho x_A + 1 - \rho, x_B), \\ T_A^f(x) &= (\rho x_A, x_B), \\ T_B^s(x) &= (x_A, \rho x_B + 1 - \rho), \\ T_B^f(x) &= (x_A, \rho x_B). \end{aligned} \tag{5}$$

Strategies

We consider stationary mixed strategies that depend only on the seller's own reputation. For each seller $j \in \{A, B\}$, a strategy is a mapping

$$\sigma_j : [0, 1] \to \Delta(\{1, 2\}), \tag{6}$$

where $\sigma_j(i \mid x_j)$ denotes the probability that seller j chooses product i when selected at reputation x_j.

Distribution-Based Oblivious Equilibrium

Suppose that seller A assumes that its competito's reputation follows a stationary distribution μ_B on $[0, 1]$. Seller A computes its expected demand as

$$d_j(x_A; \mu_B) = \int_0^1 D(x_A, y)\, \mu_B(dy) \tag{7}$$

The seller's problem becomes

$$\begin{aligned} U_A(x_A) = \gamma[1 - d_A(x_A; \mu_B)]U_A(x_A) \\ + d_A(x_A; \mu_B) \max_{i \in \{1,2\}} \Big\{ R - c_i \\ + \gamma \left[p_i U_A(T_\rho^s(x_A)) + (1 - p_i) U_A(T_\rho^f(x_A)) \right] \Big\}. \end{aligned} \tag{8}$$

Distribution-Based Oblivious Equilibrium

- Given the conjectured distribution μ_B, each seller solves its Bellman equation and obtains an optimal strategy σ.
- We require this distribution to be consistent with the reputations generated when both sellers follow σ.

Joint transition probability kernel

We define the joint transition probability kernel as

$$P^{\sigma}(x, C) = \mathbb{P}(x_{t+1} \in C \mid x_t = x, \sigma),$$

which is equivalent to

$$\begin{aligned} P^{\sigma}(x, C) = D(x_A, x_B) \Big[& p_{\sigma_A}(x_A) \mathbf{1}\{ T_A^s(x) \in C \} \\ & + (1 - p_{\sigma_A}(x_A)) \mathbf{1}\{ T_A^f(x) \in C \} \Big] \\ + [1 - D(x_A, x_B)] \Big[& p_{\sigma_B}(x_B) \mathbf{1}\{ T_B^s(x) \in C \} \\ & + (1 - p_{\sigma_B}(x_B)) \mathbf{1}\{ T_B^f(x) \in C \} \Big]. \end{aligned} \tag{9}$$

Distribution-Based Oblivious Equilibrium

A symmetric oblivious equilibrium is a triple (σ^*, μ^*, ν^*) satisfying:

1. Optimality

$$\operatorname{supp} \sigma^*(\cdot \mid x) \subseteq \arg \max_{i \in \{1,2\}} \Big\{ R - c_i + \gamma \big[p_i U_{\mu^*}(T_\rho^s(x)) + (1 - p_i) U_{\mu^*}(T_\rho^f(x)) \big] \Big\}, \quad \forall x \in [0,1].$$

2. Consistency

$$\nu^* = \nu^* P^{\sigma^*},$$
$$\mu^*(C) = \nu^*(C \times [0,1]) = \nu^*([0,1] \times C), \quad \forall C \subseteq [0,1] \text{ measurable.}$$

Mean-Draw Oblivious Equilibrium

Suppose that seller A represents its competitor by the long-run average of seller B's Thompson sampling draws:

$$p_{B,0} = \lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \widetilde{\theta}_{B,t}. \tag{10}$$

Under stationarity and ergodicity, the time average coincides with the stationary expectation almost surely. Since

$$\mathbb{E}[\widetilde{\theta}_{B,t} \mid x_{B,t}] = \frac{1-\rho+x_{B,t}}{3-2\rho}, \tag{11}$$

we obtain

$$\begin{aligned} p_{B,0} &= \mathbb{E}_{\nu}[\widetilde{\theta}_B] \\ &= \int_{[0,1]^2} \frac{1-\rho+x_B}{3-2\rho}\, \nu(dx_A, dx_B). \end{aligned} \tag{12}$$

Mean-Draw Oblivious Equilibrium

Seller A computes its expected demand as

$$d_j(x_A; p_{B,0}) = \mathbb{P}(\tilde{\theta}_A > p_{B,0} \mid x_A) \tag{13}$$

The seller's problem becomes

$$\begin{aligned} U_{p_{0,j}}(x_j) = \gamma[1 - d(x_j; p_{0,j})]\, U_{p_{0,j}}(x_j) \\ + d(x_j; p_{0,j}) \max_{i \in \{1,2\}} \Big\{ R - c_i \\ + \gamma \left[p_i\, U_{p_{0,j}}(T^s_\rho(x_j)) + (1 - p_i)\, U_{p_{0,j}}(T^f_\rho(x_j)) \right] \Big\}. \end{aligned} \tag{14}$$

Mean-Draw Oblivious Equilibrium

A symmetric oblivious equilibrium is a triple (σ^*, p_0^*, ν^*) satisfying:

1. Optimality

$$\operatorname{supp}\sigma^*(\cdot \mid x) \subseteq \arg\max_{i\in\{1,2\}} \Big\{ R - c_i + \gamma\big[p_i U_{p_0^*}(T_\rho^s(x)) + (1-p_i)U_{p_0^*}(T_\rho^f(x))\big]\Big\}, \quad \forall x \in [0,1].$$

2. Consistency

$$\nu^* = \nu^* P^{\sigma^*},$$

$$p_0^* = \int_{[0,1]^2} \frac{1-\rho+x_j}{3-2\rho}\, \nu^*(dx_A, dx_B), \quad j \in \{A, B\}.$$

Mean-Draw: Temporal Consistency

The optimality condition is unchanged. For a symmetric profile σ^*, consistency uses the expected occupation law over a finite window:

$$\bar{\nu}_{b,T}^{\sigma^*} = \frac{1}{T} \sum_{t=b}^{b+T-1} \nu_0 \left(P^{\sigma^*} \right)^t,$$

$$p_0^* = \int_{[0,1]^2} \frac{1 - \rho + x_j}{3 - 2\rho} \, \bar{\nu}_{b,T}^{\sigma^*}(dx_A, dx_B), \quad j \in \{A, B\}.$$

Experiment

$$\nu_0 = \delta_{(0.5,\,0.5)}, \qquad b = 10\,000, \qquad T = 100\,000.$$

States are measured before selection. We discard the first b periods and average the next T periods.

The average law $\bar{\nu}_{b,T}^{\sigma^*}$ need not be stationary.

Simulations

go to vscode, there are too many plots